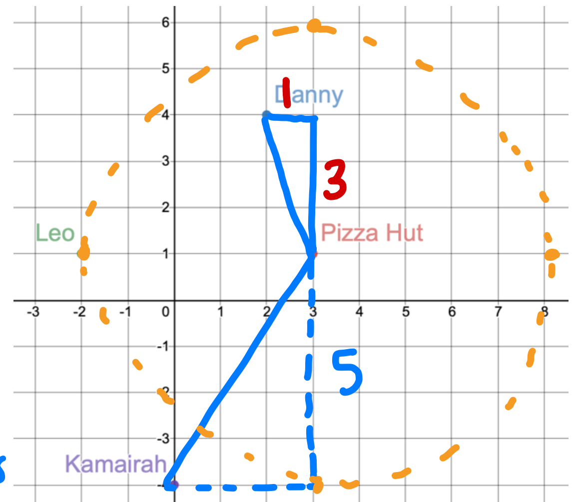
Key

Will Pizza Hut Deliver to Your House?

You are craving a pepperoni pizza for dinner and want to get it delivered to your house. When you go on Pizza Hut's website, you learn that they will deliver to locations within a 5-mile radius. Use the map to answer the questions below. Assume that each unit represents one mile and that drivers always take the shortest, most direct route between places.



1. What does it mean that Pizza Hut delivers within a 5-mile radius? How could you represent this on the graph?

They will drive up to 5 miles in any direction from Pizza Hut to deliver.
All locations w/in a circle of radius 5.

2. Will Pizza Hut deliver to Danny's house? How do you know?

yes, Danny's house is less than 5 miles away

$1^2+3^2=d^2$

$10=d^2 \qquad d=\sqrt{10}=3.16$ miles

3. Prove algebraically that Pizza Hut will NOT deliver to Kamairah's house.

$5^2+3^2=d^2$

$34=d^2 \qquad d=5.83$

since $5.83>5$, Pizza Hut won't deliver.

the distance between a pt. on the circle and the center is exactly one radius.

4. We can talk about people living within the circle, *on* the circle, or outside the Pizza Hut circle. Identify the person whose house is *on* the Pizza Hut circle. What does this mean?

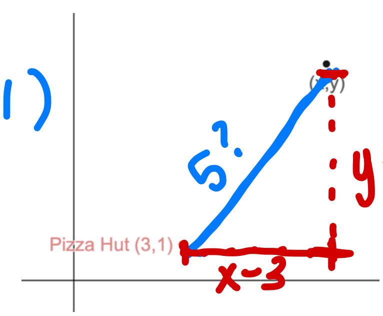
Leo is on the circle because he lives exactly 5 miles away from Pizza Hut.

5. If you knew that a person lived at some ordered pair (x, y), how could you determine if they lived *on* the Pizza Hut Circle?

Calculate the distance between (x,y) and $(3,1)$ and see if it's 5.

$\sqrt{(x-3)^2+(y-1)^2} \overset{?}{=} 5$

$(x-3)^2+(y-1)^2 \overset{?}{=} 25$



$(x-h)^2+(y-k)^2=r^2$

$(h,k)=$center

$r=$radius

6. Jimmy John's also delivers. You can figure out if Jimmy John's will deliver to your house by taking the coordinates of your house (x, y) and substituting it into the expression $(x-6)^2+(y-9)^2$. If your answer is less than or equal to 49, they will deliver to your house. Where is Jimmy John's located? What is its delivery radius?

Jimmy Johns is located at $(6,9)$ and delivers within a 7-mile radius

Lesson 0.2 – Equations of Circles

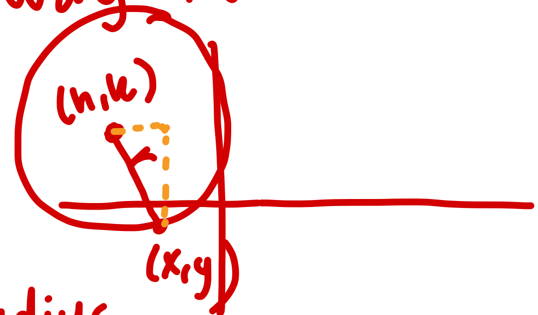
QuickNotes

Definition: A circle is the set of all points exactly one radius away from a center.

Equation of circle:

$(x-h)^2 + (y-k)^2 = r^2$

(h,k) = center r = radius



Check Your Understanding

1. Find the equation of a circle with radius 4, centered at (2, -5).

$(x-2)^2 + (y-(-5))^2 = 4^2$ or $(x-2)^2 + (y+5)^2 = 16$

2. The circle given by $(x+9)^2 + (y-1)^2 = 36$ is shifted 3 units to the right and 4 units down and its radius is doubled. Give the center and radius of the new circle.

Old center: $(-9, 1)$ Old radius: $\sqrt{36} = 6$
New center: $(-6, -3)$ New radius = 12

3. Is $(3, -2)$ inside, on, or outside the circle given by $(x-1)^2 + (y+2)^2 = 16$? How do you know?

$(3-1)^2 + (-2+2)^2 = 2^2 + 0^2 = 4$, not 16!
$(3,-2)$ is $\sqrt{4} = 2$ units away from the center, not $\sqrt{16} = 4$. Thus the pt is w/in the circle.

4. Write the equation of a circle that is centered at (3,8) and passes through $\left(\frac{1}{2}, 14\right)$.

Radius = distance between $(3,8)$ and $\left(\frac{1}{2}, 14\right)$

$= \sqrt{\left(\frac{5}{2}\right)^2 + (6)^2} = \sqrt{\frac{25}{4} + 36} = \sqrt{\frac{169}{4}} = \frac{13}{2}$

$(x-3)^2 + (y-8)^2 = \frac{169}{4}$